

LANCELIN, Australia

WMO: 956060

Lat: 31.016S

Lon: 115.330E

Elev: 16

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	44.4	46.6	32.8	27.3	63.5	35.8	30.9	60.3	32.6	60.7	30.1	61.4	5.9	90	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	15.7	93.0	68.7	88.7	68.5	85.3	68.6	75.6	83.7	73.9	81.7	72.5	80.2	11.7	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	123.1	80.7	71.4	115.9	78.8	69.8	109.8	77.3	39.2	84.2	37.5	81.5	36.2	80.3	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.8	22.7	20.8	40.3	103.9	1.7	4.0	39.1	106.7	38.1	109.1	37.1	111.3	35.9	114.2		
(5)			38.5	78.2	1.8	3.7	37.2	80.9	36.1	83.0	35.1	85.1	33.8	87.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.9	73.2	74.4	72.7	68.3	64.3	60.4	58.6	58.7	60.0	62.8	67.5	70.8		
	DBStd	7.22	5.14	5.52	5.59	4.57	4.19	3.57	3.68	3.11	3.32	4.28	4.47	5.37		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	892	2	0	3	15	63	142	198	195	154	95	20	5		
	CDD50	5816	721	683	704	550	443	312	268	271	299	396	525	644		
	CDD65	1233	257	264	243	115	40	4	1	1	3	26	95	184		
	CDH74	7693	1659	2024	1673	529	103	4	1	1	12	136	461	1090		
	CDH80	2431	539	705	536	113	11	0	0	0	1	34	132	360		
(14) Wind	WSAvg	10.5	11.8	11.2	10.5	9.0	9.0	9.9	10.2	9.8	10.8	10.8	11.4	12.0		
(15) Precipitation	PrecAvg	23.6	0.1	0.4	0.6	1.2	3.5	5.0	4.9	3.7	2.2	1.2	0.8	0.2		
	PrecMax	29.7	1.1	3.7	2.7	4.9	8.5	8.6	8.0	7.4	4.6	3.4	2.8	1.3		
	PrecMin	17.4	0.0	0.0	0.0	0.0	0.6	1.5	1.9	0.7	0.5	0.2	0.0	0.0		
	PrecStd	3.6	0.2	0.8	0.6	0.9	1.9	1.8	1.6	1.2	1.0	0.7	0.7	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.1	98.2	96.8	88.6	81.2	73.8	71.7	71.8	75.7	84.6	89.7	96.5		
		MCWB	70.7	70.6	69.4	66.1	62.3	59.7	61.4	61.6	61.1	63.0	64.5	67.5		
	2%	DB	91.1	91.7	89.4	83.1	76.7	70.8	68.2	68.9	70.9	77.2	83.5	88.2		
		MCWB	69.3	69.9	68.4	65.2	63.7	60.2	60.0	60.6	61.3	63.6	64.6	67.1		
	5%	DB	85.5	87.5	85.4	79.2	74.0	68.7	66.4	67.0	68.8	73.3	78.9	83.3		
		MCWB	69.9	70.1	68.9	65.9	63.3	60.0	59.6	60.1	61.0	63.3	64.8	67.3		
	10%	DB	81.7	83.9	81.9	76.6	71.8	67.1	65.0	65.4	66.9	70.5	75.3	79.3		
		MCWB	70.3	70.5	68.7	66.1	63.0	59.6	58.9	59.1	60.2	62.3	65.1	67.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.5	78.8	76.9	73.6	69.3	65.6	65.2	65.6	65.7	69.3	70.9	73.9		
		MCDB	86.1	86.9	84.6	78.1	73.9	69.3	68.1	69.1	71.2	75.5	78.0	82.4		
	2%	WB	75.2	75.9	74.1	71.2	67.5	63.8	62.7	63.4	64.0	66.6	69.1	72.0		
		MCDB	83.4	84.1	82.2	76.5	72.5	67.9	66.1	67.0	68.7	72.6	76.5	80.9		
	5%	WB	73.2	74.1	72.5	69.6	65.9	62.4	61.3	61.8	62.4	65.0	67.6	70.5		
		MCDB	80.6	82.4	80.4	75.4	71.4	66.9	65.1	65.5	67.3	70.8	75.0	78.9		
	10%	WB	71.4	72.5	71.0	68.0	64.4	61.0	60.0	60.2	61.0	63.5	66.1	69.0		
		MCDB	78.9	80.6	78.7	73.8	70.2	65.7	64.1	64.3	65.9	69.3	73.2	77.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.0	15.7	16.7	15.9	15.2	13.9	12.9	13.7	13.6	15.1	14.4	14.1		
		MCDBR	24.1	22.8	23.1	20.6	18.5	16.9	15.0	17.1	18.4	21.6	23.6	22.9		
	5% WB	MCWBR	12.9	11.3	12.6	12.6	11.9	10.3	10.2	11.6	12.3	13.3	13.3	12.3		
		MCWBR	16.8	16.3	17.0	15.1	14.3	12.7	11.5	13.9	15.0	16.8	16.8	17.3		
(40) Clear-Sky Solar Irradiance	taub		0.347	0.346	0.332	0.321	0.310	0.301	0.301	0.309	0.328	0.328	0.330	0.335		
		taud	2.557	2.578	2.603	2.624	2.620	2.630	2.620	2.591	2.517	2.540	2.556	2.560		
	Ebn at Noon		315	309	303	291	279	274	279	291	299	312	318	319		
		Edh at Noon	34	33	30	26	24	22	23	27	32	34	34	35		
(44) All-Sky Solar Radiation	RadAvg		2613	2371	1968	1465	1109	916	966	1246	1652	2124	2497	2683		
	RadStd		123	75	99	79	71	61	57	53	87	84	82	105		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (14 neighbors)	N/A	N/A	N/A	+0.99	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page